

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	2 MARKS

3.2 – Solution Requirement

The solution for India's Agricultural Crop Production Analysis using Tableau is based on a plan. This plan is designed to change data into useful information. At the heart of this project is a defined problem. The agriculture domain is a part of the supply chain and is expected to change a lot.

The main goal is to use a dataset on crop production in India to predict production and find key indicators and metrics that affect it. To achieve this goal we need to follow steps. The first and most important phase is data preparation. This involves using tools like Microsoft Excel to clean the data remove duplicates, spelling and handle missing values. This ensures that the data is accurate.

Many projects also use Python for data analysis using libraries like Pandas, NumPy and Matplotlib to understand the data better check for missing values and prepare the dataset for visualization. The data usually comes from government databases, such as the Ministry of Agriculture & Farmers Welfares portal and is supplemented with additional datasets on rainfall or population from platforms like Kaggle.

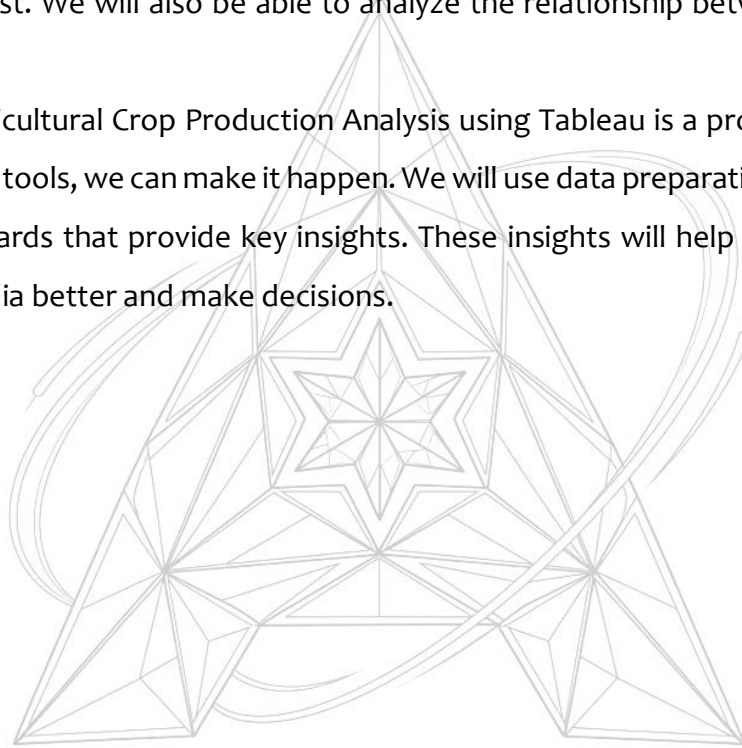
The next phase involves creating visualizations and dashboards in Tableau. This is where the analytical requirements come to life. The solution must provide a dashboard that gives key insights, such as trend analysis of production over the years comparisons of seasonal production, state-wise performance and the relationship between crop area and production.

The final product must be not just a static report. It needs features like filters, parameters and slicers to allow users to explore the data based on their needs, such as looking

at a specific state, crop or year. Creating a "story" within Tableau is also important to guide users through the findings in an compelling way.

We are using Tableau to make this happen. Tableau is a tool that helps us create interactive dashboards. The dashboards will show us how crop production in India is changing over time. We will be able to see which states are producing crops and which crops are being produced the most. We will also be able to analyze the relationship between crop area and production.

Indias Agricultural Crop Production Analysis using Tableau is a project. With the right plan and the right tools, we can make it happen. We will use data preparation and visualization to create dashboards that provide key insights. These insights will help us understand crop production, in India better and make decisions.



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